



DESIGN AND ANALYSIS OF ALGORITHMS

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DESIGN AND ANALYSIS OF ALGORITHMS

Sequential Search

Major Slides Content: Anany Levitin

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DESIGN AND ANALYSIS OF ALGORITHMS

Sequential Search

- Compares successive elements of a given list with a given search key until:
 - A match is encountered (Successful Search)
 - List is exhausted without finding a match (Unsuccessful Search)
- An improvisation to the algorithm is to append the key to the end of the list
- This means the search has to be successful always and we can eliminate the end of list check

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Sequential Search

- Sequential / Linear Search

10	14	19	26	27	31	33	35	42	44
0	1	2	3	4	5	6	7	8	9

- For key = 33, 6 is returned
- For key = 50, -1 is returned

Algorithm SequentialSearch($A[0..n-1]$, K)

//Outputs the index of the **first** element of A that
// matches K or -1 if there are no matching
elements.

$i \leftarrow 0$

while ($i < n$) and ($A[i] \neq K$) do

$i \leftarrow i + 1$

if ($i < n$) return i

return -1

Input size: n .

Basic Operation: $(i < n) \text{ and } (A[i] \neq K)$

$$C_{\text{worst}}(n) = n$$

$$C_{\text{best}}(n) = 1$$

Let ' p ' be the probability of the search key present in the array.

$$C_{\text{avg}}(n) = p(n + 1) / 2 + (1 - p)n$$

Input Size: n

Basic Operation : $(i < n)$ and $(A[i] \neq K)$

$C_{\text{worst}}(n)$ = Count of the basic operation at the max
 $= n \in \Theta(n)$

$C_{\text{best}}(n) = 1 \in \Theta(1)$

$C_{\text{avg}}(n)$ = from $(n+1)/2$ to n depending on the probability of search key being present in the input array.

$C_{\text{avg}}(n) \in \Theta(n)$

ALGORITHM *SequentialSearch2*($A[0..n]$, K)

//Implements sequential search with a search key as a sentinel

//Input: An array A of n elements and a search key K

//Output: The index of the first element in $A[0..n - 1]$ whose value is

// equal to K or -1 if no such element is found

$A[n] \leftarrow K$

$i \leftarrow 0$

while $A[i] \neq K$ **do**

$i \leftarrow i + 1$

if $i < n$ **return** i

else return -1

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Brute – Force String Matching

String Matching - Terms

- pattern:
a string of m characters to search for
- text:
a (longer) string of n characters to search in
- problem:
find a substring in the text that matches the pattern

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String Matching Idea

Step 1: Align pattern at beginning of text

Step 2: Moving from left to right, compare each character of pattern to the corresponding character in text until:

- all characters are found to match (successful search); or
- a mismatch is detected

Step 3: While pattern is not found and the text is not yet exhausted, realign pattern one position to the right and repeat Step 2

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String Matching

```
ALGORITHM BruteForceStringMatch(T[0 .. n - 1], P[0 .. m - 1])
//Implements brute-force string matching
//Input: An array T[0 .. n - 1] of n characters representing a text
// and an array P[0 .. m - 1] of m characters representing a pattern
//Output: The index of the first character in the text that starts a
//matching substring or -1 if the search is unsuccessful
for i ← 0 to n-m do
    j ← 0
    while j < m and P[j] = T[i + j] do
        j ← j+1
    if j = m return i
return -1
```

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String Matching Example

T	H	I	S		I	S		A		S	I	M	P	L	E		E	X	A	M	P	L	E
S	I	M	P	L	E																		
	S	I	M	P	L	E																	
		S	I	M	P	L	E																
			S	I	M	P	L	E															
				S	I	M	P	L	E														
					S	I	M	P	L	E													
						S	I	M	P	L	E												
							S	I	M	P	L	E											
								S	I	M	P	L	E										
									S	I	M	P	L	E									
										S	I	M	P	L	E								

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String Matching Analysis

Worst Case:

- The algorithm might have to make all the 'm' comparisons for each of the $(n-m+1)$ tries
- Therefore, the algorithm makes $m(n-m+1)$ comparisons
- Brute Force String Matching is a $O(nm)$ algorithm

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Sequential Search - Multiple choice questions

Consider a sequential search on an unsorted array of n distinct elements. The probability that the key is present in the array is p , and if present, it is equally likely to be at any position. Which of the following correctly represents the expected number of comparisons?

- a) $(n+1)/2$
- b) $p(n+1)/2 + (1-p)n$
- c) $pn + (1-p)n/2$
- d) $pn/2$

Answer : b) $p(n+1)/2 + (1-p)n$

Design and Analysis of Algorithms

Sequential Search - Multiple choice questions

A sequential search algorithm is implemented on an array of size n using a sentinel.

Assume that: The key is not present in the array. No checking for the bounds as the n th value would contain the key.

Which of the following statements is most accurate regarding the number of comparisons performed?

- a) Exactly n comparisons are performed.
- b) Exactly $n+1$ comparisons are performed.
- c) At most n comparisons are performed, and the asymptotic time complexity remains $\Theta(n)$
- d) The number of comparisons is reduced to $\Theta(\log n)$ due to elimination of boundary checks.

Answer **b)** Exactly $n+1$ comparisons are performed.

Design and Analysis of Algorithms

Sequential Search - Multiple choice questions

A sequential search is performed on an array of size n .

Assume that: The key is always present in the array. The probability that the key is present is the same for every position.

What is the **expected number of comparisons** performed by the sequential search?

- a) $(n+1)/2$
- b) $2(n+1)/3$
- c) $n(n+1)/2$
- d) $(3n+1)/2$

Answer : a) $n+1/2$

Design and Analysis of Algorithms

String Matching - Multiple choice questions

Consider the brute force string matching algorithm used to find a pattern P of length m in a text T of length n ($n \geq m$)

Which of the following expressions correctly represents the worst-case number of character comparisons?

- a) $n-m+1$
- b) $n+m$
- c) $m(n-m+1)$
- d) $n \log m n$

Answer : **c)** $m(n-m+1)$

Design and Analysis of Algorithms

String Matching - Multiple choice questions

Let Text $T=AAAAAA$ (length $n=6$) Pattern $P=AAA$ (length $m=3$)

The brute force string matching algorithm compares characters from left to right at each shift and reports all valid occurrences of the pattern. How many character comparisons are performed in total?

- a) 9
- b) 10
- c) 12
- d) 15

Answer C : 12

Design and Analysis of Algorithms

String Matching - Multiple choice questions

The brute force string matching algorithm compares the pattern from left to right at every shift. Which of the following text–pattern pairs will result in the worst-case number of character comparisons?

- a) Text : ABCDEFABCDEF , Pattern : DEF
- b) Text : AAAAAAAAAA , Pattern : AAAA
- c) Text : AAAAAAAAAA , Pattern : AAAAB
- d) Text : ABABABABAB , Pattern : ABAB

Answer : c) Text : AAAAAAAAAA , Pattern : AAAAB



THANK YOU

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